TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Analysis of Group IIIB (Co, Ni, Zn, Mn)

The ammoniacal-sulphide group — white, flesh and black sulphides sorted by dissolution behaviour, with the confirmatory tests for cobalt, nickel, zinc and manganese.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 8 · LEVEL 1 TERM 1

1. Aim and first reading

Aim: to analyse the Group-IIIB precipitate — the sulphides thrown down by H_2S from the ammoniacal filtrate (Experiment 4) — and identify Co^{2+}, Ni^{2+}, Zn^{2+} and Mn^{2+}. Colour divides the group at sight:

SULPHIDE	COLOUR
ZnS	White
MnS	Flesh-coloured / pale pink-buff
CoS, NiS	Black

A white precipitate in this group can only be zinc; a flesh-coloured one, manganese. The black pair (Co, Ni) share colour and must be parted by the tests of §3, which is also where their famous individual confirmations live.

2. Separation outline

1. Treat the washed precipitate with **cold dilute HCl**: ZnS and MnS dissolve; CoS and NiS (once precipitated and aged) do not dissolve in dilute acid — filter.
   Filtrate (Zn^{2+}, Mn^{2+}): boil off H_2S, make alkaline with NaOH in excess — $Zn(OH)_2$ dissolves (amphoteric) while $Mn(OH)_2$ precipitates and browns in air. Filter again: residue → manganese (§3); filtrate → zinc (§3).
2. **Residue** (CoS + NiS): dissolve in aqua regia, evaporate nearly to dryness, take up in water, and test separate portions for cobalt and nickel (§3).

3. Confirmatory tests

CATION	CONFIRMATIONS
Co^{2+}	(i) Borax bead: deep blue bead in both flames — beautiful and specific. (ii) Solution + ammonium thiocyanate in amyl alcohol (or acetone): blue colour in the organic layer ($[Co(SCN)_4]^{2-}$). (iii) KNO_2 in acetic acid gives a yellow precipitate (potassium cobaltinitrite) on standing.
Ni^{2+}	Dimethylglyoxime (DMG) in slightly ammoniacal solution → bright red (scarlet) precipitate — among the most famous spot tests in analysis, and effectively specific to nickel here.
Zn^{2+}	From the aluminate-style filtrate (zincate): pass H_2S → white ZnS again; or acidify slightly and add potassium ferrocyanide → white precipitate. NaOH behaviour (white ppt soluble in excess) supports.
Mn^{2+}	The hydroxide precipitate browns in air (oxidation to hydrated MnO_2). Strong confirmation: fuse/boil a portion with PbO_2 (or $NaBiO_3$) in concentrated HNO_3 → purple colour of permanganate — manganese oxidised all the way to MnO_4^- announces itself unmistakably.

4. Notes

- Freshly precipitated CoS/NiS dissolve in dilute acid more readily than aged ones — the acid treatment must be cold and brief, and the separation checked, or cobalt/nickel leak into the Zn/Mn fraction.
- MnS's flesh colour is delicate; in mixtures the white of ZnS and black of CoS/NiS easily mask it — the NaOH/air-browning behaviour is the practical finder.
- DMG test solutions must be only faintly ammoniacal; strong ammonia and large amounts of cobalt both disturb the red precipitate's formation.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A IIIB precipitate is flesh-coloured. Name the sulphide and give the strong confirmatory test with its result.
2. How does cold dilute HCl divide the group, and what caution applies to the black pair?
3. State the dimethylglyoxime test — reagent condition and positive result — and the cation it confirms.
4. Give two confirmatory tests for cobalt, one of them a dry test.
5. Zn and Mn travel together into the HCl filtrate. How are they separated, and what visible change betrays manganese during it?

6. Why is a white IIIB precipitate decisive for zinc, while a black one is not decisive for either of its candidates?

Answer key

1. MnS — manganese. Confirmation: oxidise with PbO_2 or $NaBiO_3$ in concentrated HNO_3 — the solution turns purple (permanganate formed).
2. ZnS and MnS dissolve; aged CoS and NiS do not. Caution: fresh CoS/NiS are more soluble — work cold and quickly, or they contaminate the soluble fraction.
3. Dimethylglyoxime added to the faintly ammoniacal solution gives a bright scarlet-red precipitate — confirms Ni^{2+}.
4. Dry: borax bead — deep blue in oxidising and reducing flames. Wet: ammonium thiocyanate in amyl alcohol/acetone — blue $[Co(SCN)_4]^{2-}$ layer; (or KNO_2/acetic acid — yellow cobaltinitrite precipitate).
5. Add excess NaOH: zinc's hydroxide dissolves (zincate, filtrate), manganese's hydroxide precipitates — and betrays itself by browning in air as it oxidises.
6. Because within this group only ZnS is white and only MnS is flesh-coloured, while CoS and NiS share black — colour alone settles the light pair's members one-to-one but leaves the black pair to chemical separation and their specific tests (DMG for Ni, bead/thiocyanate for Co).

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